

PAPER_945: Merger Phase Lag

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** `smbh_binary_mergers.py` (MergerPhaseLag) **Calculator:** MergerPhaseLagCalc (CP4 #529) **CVW:** v2.0.0 compliant

Abstract

We derive the UQFF phonon-induced gravitational-wave phase lag for SMBH binary mergers in the LISA band (1-100 mHz). The cumulative phase shift $\Delta\Phi = 2\pi(f_{\text{max}} - f_0) \cdot D_{\text{total}}(q) \cdot S_{26}$ yields 200-400 cycles depending on mass ratio, providing a distinctive observational signature separable from GR waveform templates.

1. Phase Lag Formula

$$\Delta\Phi = 2\pi(f_{\text{max}} - f_0) \cdot D_{\text{total}}(q) \cdot S_{26}$$

where $S_{26} = \sum_{k=1}^{26} e^{-[\text{SSq}] \cdot k/26}$ with $[\text{SSq}] = 0.57$.

2. Phase Lag vs Mass Ratio

q	D_{total}	$\Delta\Phi$ (rad)	Cycles
0.2	0.491	~ 1900	~ 302
0.5	0.432	~ 1670	~ 266
0.8	0.372	~ 1440	~ 229
1.0	0.333	~ 1290	~ 205

3. Detectability

At LISA sensitivity ($\delta\Phi \sim 0.1$ rad), phase lags of ~ 200 -400 cycles are detectable with $\text{SNR} > 10^3$, making this the most constraining UQFF prediction for space-based GW detectors.

4. Source Data

- **File:** `smbh_binary_mergers.py`
 - **Session:** 213
 - **CP4 Class:** MergerPhaseLagCalc (#529)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Berti, E. et al. (2006) – PRD, 73, 064030
3. LISA Consortium (2023) – arXiv:2402.07571